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APPLICATION

Rthoptera: Standardised insect bioacoustics in R

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Abstract

1. Almost 100 years after the dawn of insect bioacoustics, freely available and robust analysis tools and protocols are still needed.
2. We introduce *Rthoptera*, an open-source R package designed for robust analysis of insect acoustic signals. It delivers accurate temporal and spectral measurements, along with intuitive visualisations, through a streamlined workflow. Interactive applications ensure accessibility for researchers across all levels of technical expertise.
3. Four new acoustic metrics for discrete units are introduced: Spectral Excursion, Pattern Complexity Index, Temporal Excursions and Dynamic Excursion. A Broadband Activity Index with potential use in pest monitoring is also showcased.
4. Accompanied by an appropriate recording protocol, our workflow can become part of a standard analysis protocol for insect bioacoustics.

KEYWORDS
acoustics, automated measurements, interactive plots, orthoptera, R-package

1 | INTRODUCTION

The current rate of species loss is outpacing our ability to track biodiversity change, highlighting the need for improved monitoring systems (O'Connor et al., 2020). Addressing this challenge, passive acoustic monitoring is helping researchers track soniferous fauna at the community and ecosystem scales (Sugai et al., 2019), with at least two approaches leveraging this technology. The soundscape ecology/ecoacoustics approach focusses on broad acoustic patterns reflecting the response of animal communities to environmental change (Pijanowski et al., 2011; Pijanowski & Rivas-Fuenzalida, 2024) often using free software tools such as *soundecology* (Villanueva-Rivera et al., 2018) and *scikit maad* (Ulloa et al., 2021). Contrasting with the soundscape focus, the bioacoustics approach relies heavily on focalised recordings for detailed analysis, focussing on the behavioural, ecological and evolutionary implications of animal acoustic communication. For biodiversity monitoring and conservation purposes, bioacoustic studies characterise signals ('acoustic keys') to identify species in the field and in recordings, which are then manually annotated with free software, for example *Whombat* (Balvanera et al., 2023), and used as training data for automated detectors with the help of machine learning tools (Hafner & Katz, 2018; Pijanowski et al., 2024). Although automatic identification has seen tremendous progress in birds (e.g.

The source code for this package can be found on GitHub: https://github.com/naturewaves/Rthoptera. Documentation and vignettes can be explored on the website: https://naturewaves.github.io/Rthoptera/index.html.

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BirdNet (Kahl et al., 2021), *AviaNZ* (Marsland et al., 2019)) and bats (Roemer et al., 2021), with several broad-scope tools available (e.g. *OpenSoundscape* (Lapp et al., 2023), *ARBIMON* (Aide et al., 2013), *Koogu* (Madhusudhana, 2023), *Ketos* (Kirsebom et al., 2021)), its development in insects has been slower, especially in the Global South (Ragge & Reynolds, 1998; Riede & Balakrishnan, 2024), where the required acoustic keys are not always available due to lack of training in bioacoustics and/or insufficient resources for fieldwork. This emphasises the need for promoting new communities of practice and providing accessible tools for descriptive insect bioacoustics, which would boost research in several fields, from agricultural pest monitoring (Mankin et al., 2021), behavioural (Pollack, 2017) and evolutionary (Desutter-Grandcolas, 2003; Otte, 1992; Robillard et al., 2007) studies, through monitoring terrestrial and freshwater ecosystems (Aiken, 1985; Greenhalgh et al., 2021; Linke et al., 2018) affected by climate change and habitat loss.

Insect bioacousticians without access to expensive dedicated software rely on multi-software approaches to extract measurements and graphs, often resulting in time-consuming workflows. While free versions of bioacoustic software exist, they are typically optimised for analysing frequency-modulated signals, such as those produced by mammals and birds, for example Raven (Charif et al., 2010), and many tools lack flexibility due to their proprietary nature. A popular alternative is to use open-source software such as R (Ihaka & Gentleman, 1996), allowing customised analyses while requiring programming skills, which prevents its widespread adoption. This can be addressed by embedding functions into graphical user interfaces (GUIs).

Here, we introduce *Rthoptera*, a Shiny-powered (Jia et al., 2022) R package designed as an intuitive workflow for the extraction of taxonomically relevant spectral and temporal statistics from audio recordings of insect sounds and the production of standardised graphs commonly used in literature of this taxon. The workflow is structured in three sections: (1) import and preprocessing, (2) analysis and (3) visualisation. The last two sections are specifically tailored for insect bioacoustics with a semi-automated approach aimed at minimising human bias, returning results faster than 'manual' methods using a multi-software approach (https://naturewaves.github.io/Rthoptera/articles/Compare.html). Our main goals are (1) encouraging the production of high-quality analyses by researchers without requiring extensive expertise in R or bioacoustics and (2) providing a reproducible, standardised and interactive workflow for producing graphs and measurements of insect acoustic signals. We foresee that the most common use cases for our package will include the extraction of acoustic features to support studies in taxonomy, behaviour and evolution, and the automatic detection of species, including real-time monitoring of insect pests, and the characterisation of *acoustic types*, *sonotypes* and *ethospecies*. We define *acoustic type* as a focalised, high signal-to-noise recording of the calling song, which has been selected to serve as the main reference for the acoustic identification of a species. Evidently, linking an *acoustic type* to the type specimen is only possible for newly described species for which recordings are available, whereas for most species, the *acoustic type* is recorded from a non-type specimen, which we call the 'acoustic voucher specimen'. This concept differs from *ethospecies* (Riede, 1993), *acoustic morphospecies* (Aide et al., 2017) and *sonotype*, which refer broadly to a stereotypical acoustic pattern, often not assigned to a particular species, used to quantify species richness through an agnostic approach. Crucially, an *acoustic type* recording must be backed by a catalogued voucher specimen.

The latest version of *Rthoptera* can be freely downloaded and installed directly from *R* or the *RStudio* interface using:

```
install.packages("remotes")
remotes::install_github("naturewaves/Rthoptera")
library(Rthoptera)
```

For a complete list and description of the regular functions (not apps) in the package, visit:

https://naturewaves.github.io/Rthoptera/reference/index.html

2 | PACKAGE DEVELOPMENT AND FEATURES

The names of regular R functions are presented in monospaced font followed by parentheses (e.g. function_name()), Shiny apps are displayed in monospaced font without parentheses (e.g. app_name), and R packages are italicised (e.g. *packageName*). After consolidating the main visualisation and data analysis techniques used by researchers in insect bioacoustics, we built upon routines from *tuneR* (Ligges et al., 2023) and *seewave* (Sueur et al., 2008) to create new visualisation and analysis functions. We also included interactive versions of common import and editing functions from these packages (see Section 3). Similarly, although most of the spectral measurements are also found in other packages (e.g. *warbleR*, which requires importing selection tables created in Raven), our interactive plots illustrating the measurements can be more intuitive to both early career researchers and end users of the published materials, as these can be saved and made available as HTML files. Static plots in the package are built with *ggplot2* (Wickham, 2016), while interactive plots use *plotly* (Sievert et al., 2021). Moreover, to our knowledge, the automatic temporal measurements based on peak count for low-Q songs and including several new metrics specifically designed for the characterisation of insect acoustic signals are not available in any other package.

After developing the functions, we used ChatGPT-4 (OpenAI, 2024) as an assistant for coding the *Shiny* applications and developing the documentation. Each code chunk generated in this way was manually analysed and debugged. The package was tested with *usethis* (Wickham et al., 2024) and *devtools* (Wickham et al., 2022), reaching 98% coverage, while operating system tests for Windows, MacOS and Linux were successfully completed with *rhub* (Csárdi & Salmon, 2024).

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Rthoptera offers 17 Shiny applications, accessible with the launch_app() function (e.g. launch_app("spectrum")). For a detailed guide on how to launch them, visit https://naturewaves.github.io/Rthoptera/articles/Apps.html. All the analysis apps have regular function versions, allowing users to work from the console or a script if they prefer.

A summary of the applications can be found in Table 1.

2.1 | Analysis formats

Insect acoustic signals are usually measured and visualised in three basic analysis formats:

- *Oscillogram*: A waveform representing changes in the amplitude (energy) of a sound over time. All temporal measurements in

TABLE 1 Summary of the apps in Rthoptera.

App	Workflow	Purpose	Output
import_wave[a]	Preprocessing	Browse and import Waves into R	Wave
trim[a]	Preprocessing	Select and trim Waves	Wave
bandpass_filter[a]	Preprocessing	Visualise Wave and apply HPF and/or LPF	Wave
downsample[a]	Preprocessing	Visualise Wave and reduce sampling rate	Wave
merge_waves[a]	Preprocessing	Concatenate several Waves horizontally (in time)	Wave
compose_waves[a]	Preprocessing	Merge and organise several Waves horizontally (in time) with recycling	Wave
song_stats_lq	Analysis	Automatically extract temporal and spectral statistics. Optimised for broadband sounds	HTML, CSV
song_stats_hq	Analysis	Automatically extract temporal and spectral statistics. Optimised for narrowband sounds	HTML, CSV
spectral_stats	Analysis	Extract spectral statistics	HTML, CSV
Spectrogram	Visualisation	Produce a standard static spectrogram (automatic window length)	PNG
multi_plot	Visualisation	Produce an aligned, static spectrogram + mean power spectrum + oscillogram plot	PNG
multi_meanspectra_ggplot[a]	Visualisation	Produce a static colour-coded oscillogram and corresponding mean power spectra	PNG, HTML
multi_meanspectra_plotly[a]	Visualisation	Produce an interactive colour-coded oscillogram and corresponding mean power spectra	PNG, HTML
spectrum[a]	Visualisation	Produce a static power spectrum (mean, SD, etc.)	PNG
oscillogram_zoom[a]	Visualisation	Produce a stack of up to 3 zoomed oscillograms	PNG
colorcode_oscillogram[a]	Visualisation	Produce a colour-coded oscillogram from interactive selections	PNG
multi_oscillogram[a]	Visualisation	Produce a stack with oscillograms from different recordings	PNG

[a]Available only as app.

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Rthoptera are calculated on the envelope of the waveform, which is obtained with the env() function from *seewave*.

- *Mean Power Spectrum*: A graph representing the acoustic power distribution across frequencies. The underlying array containing the actual data for the plot informs all spectral measurements in *Rthoptera*. While other software allows visualising the mean spectrum from 'point' selections (i.e. 'instantaneous spectrum'), our package avoids this approach by design to minimise bias.
- *Spectrogram*: Also known as 'sonogram'. A grey-scale or colour-scale plot, which is the result of a Short-Time Fourier Transform (STFT) applied to the waveform, showing frequency on the *Y*-axis, time on the *X*-axis and energy on the *Z*-axis (i.e. pixel intensity). While this plot is often used for visualisation purposes and to calculate some spectral acoustic indices (Sueur et al., 2014), extracting manual measurements from it is not recommended as it increases the likelihood of measurement bias.

2.2 | Terminology

Nomenclature inconsistencies in the insect bioacoustics literature motivated Baker and Chesmore (2020) to develop a controlled vocabulary. Some of these discrepancies are still difficult to resolve, mainly because researchers attempt to define units based on behaviour-agnostic (e.g. wave train) and behaviour-explicit terms (e.g. 'produced by a complete cycle of the forewings') simultaneously. Considering this issue, we opted for using a behaviour-agnostic terminology for the automated measurements, which can be described a posteriori with behaviour-explicit terminology.

- *Peak*: The song_stats_lq app identifies individual peaks in the envelope, usually corresponding to 'tooth impacts' or 'impulses'. See Baker and Chesmore (2020) for details.
- *Train*: A set of consecutive peaks, synonym of 'pulse' or 'syllable', depending on the context (Ragge & Reynolds, 1998).
- *Motif*: First-order aggregation of trains ('dyplosyllable', 'echeme', etc.). The hierarchy can be extended beyond this point as needed (e.g. first-order motif sequence, second-order, etc.).

3 | PREPROCESSING

Rthoptera wrapped functions from other packages into interactive Shiny applications for importing, downsampling, filtering, trimming, and merging Wave objects, with the added convenience of real-time visual feedback to support editing decisions. For guidance on their use, visit https://naturewaves.github.io/Rthoptera/articles/Preprocess.html.

4 | ANALYSIS

4.1 | Spectral statistics

The spectral_stats app (Figure 1) focusses on the overall spectral structure of a complete Wave object. It computes several spectral measurements, returning an interactive plot and a table with the results, including the parameters used. The plot can be saved as HTML. The table includes:

- *Peak Frequency*: The frequency with maximum amplitude value. Also known as 'carrier frequency'.
- *Low Frequency*: The first frequency below the selected amplitude threshold to the left of the peak frequency (with frequency on the X-axis).
- *High Frequency*: The first frequency below the selected amplitude threshold to the right of the peak frequency (with frequency on the X-axis).

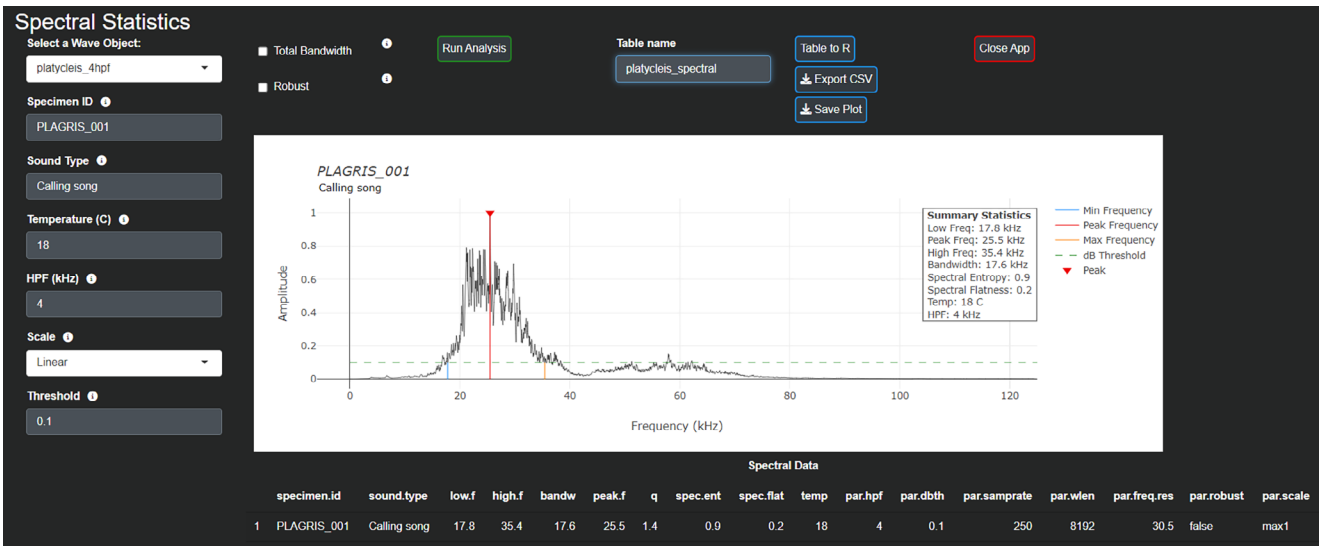

FIGURE 1 The spectral_stats app.

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- *Bandwidth*: The difference between High Frequency and Low Frequency (kHz).
- *Spectral Excursion*: The length of the spectral contour along the bandwidth (at −20 dB).
- *Spectral Energy*: The area under the curve along the bandwidth of the signal.
- *Spectral Entropy*: The Shannon entropy of the spectrum, using sh() from *seewave*.
- *Spectral Flatness*: The spectral flatness of the Wave, using sfm() from *seewave*.
- *Quality Factor* (Q): The peak frequency divided by the bandwidth at −3 dB. Incorporated experimentally in *Rthoptera* and calculated as follows:

$$Q = \frac{F_0}{\text{bandwidth at} - 3\text{dB}} \quad (1)$$

where F_0 is the frequency of maximum amplitude, while the bandwidth is calculated with the threshold set at 3 dB below the peak amplitude. See Bennet-Clark (1999) for detailed explanation and advice on its use and interpretation.

4.2 | Temporal statistics

One of the main challenges of insect bioacoustics is finding ways to objectively measure temporal features (e.g. durations of elements, syllables, echemes and gaps), especially considering the high variability of the signals due to differences in temperature. While most of the available summary metrics (e.g. Acoustic Complexity, Temporal Entropy) are useful in soundscape analysis and are structure-agnostic (i.e. they return a value for the whole duration of the sound or a user-defined time period), focalised insect recordings can be better characterised by summarising to the structure level (e.g. chirps, echemes or motifs). *Rthoptera* addresses these challenges by detecting and discretising units (e.g. trains) and unit groups (e.g. motifs) before extracting robust metrics (e.g. the Pattern Complexity Index) based on those units and their proportions, accounting for the variability in speed while conserving the temporal structure. Considering the diversity of sound production in insects, *Rthoptera* offers two functions to extract acoustic features considering the general structure of the produced sound. Following Montealegre-Z and Morris (1999), we distinguish between High-Quality Factor (HQ) and Low-Quality Factor (LQ) calls. HQ calls have a tonal, 'musical' quality, producing a narrow frequency band usually accompanied by one or several harmonics of varying intensity. HQ calls are produced by most field crickets, ground crickets and tree crickets. For these signals, it is appropriate to use a cut-off amplitude threshold to find the start and end of each sound, which is the approach used by the song_stats_hq app (Figure 2). In contrast, LQ sounds are broadband, with a harsh, 'noisy' timbre, common in most katydids, bush crickets, grasshoppers, cicadas, water-bugs and some beetles. To measure these signals, the song_stats_lq app (Figure 3) uses a 'peakfinder' algorithm to detect local maxima along the waveform, locating these 'transient' (LQ) sounds and aggregating them into trains. This allows to accurately measure those trains containing peaks that are too faint (e.g. wing opening), but which may contain relevant information. These softer trains are often below the amplitude level of the

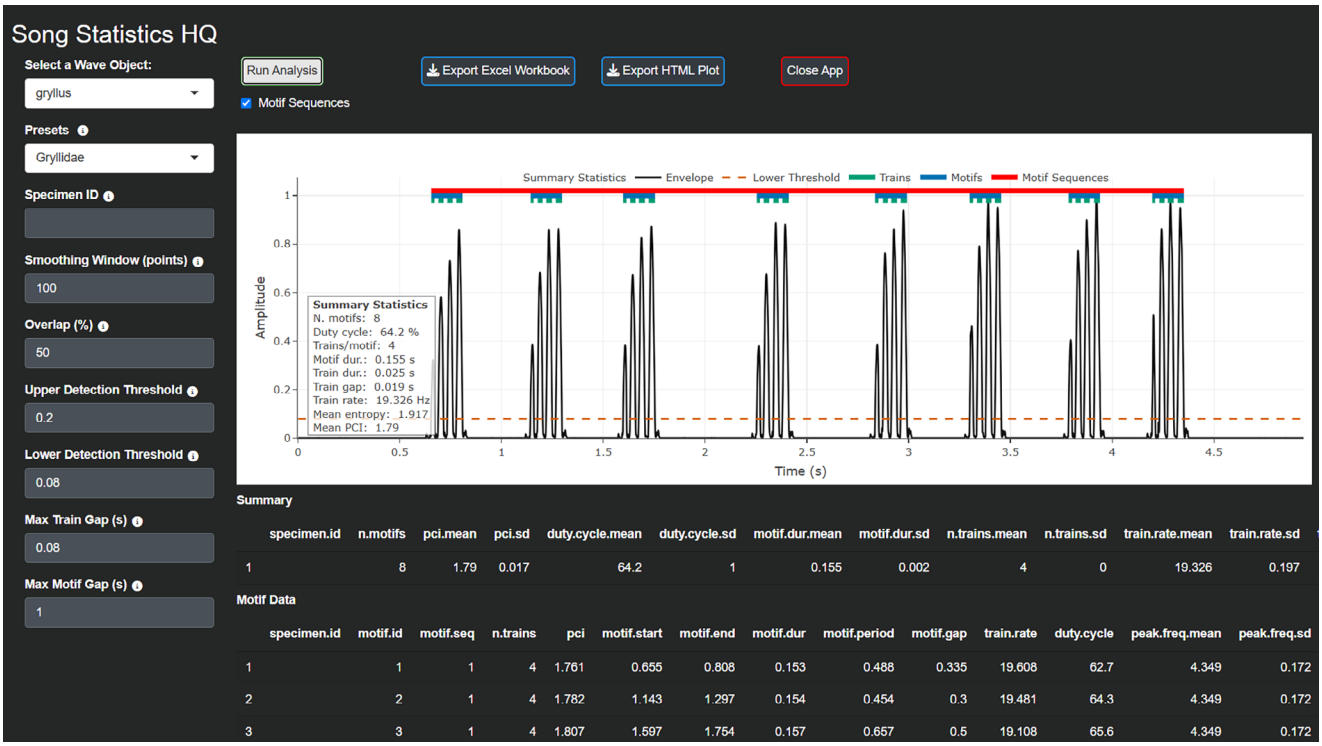

FIGURE 2 The song_stats_hq app. This function is optimised for tonal calls, like this example of *Gryllus campestris*.

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FIGURE 3 The song_stats_lq app detects local peaks (tooth impacts), aggregating them into trains for temporal and spectral measurements. This example shows the song of *Coryphoda albidicollis*, a Phaneropterinae from Chile.

TABLE 2 Suitability of each song statistics app for six taxonomic groups known to produce sounds.

Function	Tettigoniidae	Gryllidae	Acrididae	Coleoptera	Cicadidae	Nepomorpha
song_stats_lq	YES	NO[a]	YES	YES	NO[a]	YES
song_stats_hq	NO[a]	YES	NO[a]	NO[a]	YES[a]	NO[a]

[a]Not optimised for this taxon, potentially useful with appropriate parameters.

reverberations of the loudest—often terminal—parts of the call and therefore are harder to capture with the amplitude threshold approach. Once the individual trains are detected, a set of temporal and spectral metrics are calculated for each detection. Then, based on the parameters selected by the user, it aggregates the trains into motifs and, optionally, motif sequences, providing summary statistics in separate tables for each. See Table 2 for guidance on the suitability of these two apps for the main soniferous taxonomic groups.

Rthoptera introduces three temporal metrics: the Pattern Complexity Index (PCI), Temporal Excursion (TE) and Dynamic Excursion (DE). The PCI aims to characterise the temporal structure of a motif (e.g. an echeme) with a procedure that is robust to the variability caused by differences in temperature. First, we take the proportions of the duration of trains and gaps across the motif and then calculate PCI as follows:

$$\mathrm{PCI} = \frac{H \times CV + \sqrt{n}}{\sqrt{d} + 1} \tag{2}$$

where H is the Shannon entropy of the proportions, CV is the coefficient of variation of the proportions, n is the number of trains in the motif and d is the duration of the motif.

The TE and DE metrics evaluate temporal variability small analytical units (e.g. a train or syllable) of LQ calls in peak rate and amplitude, respectively, increasing the potential for studying acoustic variability at individual and population scales and experimentally testing the effects of external factors on song features.

TE takes the absolute differences between consecutive peak periods in the train and adds them together. Consequently, a train containing regularly spaced peaks will return a low TE value. TE is calculated as follows:

$$\mathrm{TE} = \sum_{i=2}^{n} |P_i - P_{i-1}| \tag{3}$$

where P_i represents the i-th peak period, in milliseconds, and n is the total number of peaks for a given train. Missing values are ignored in the summation. Along with PCI, measures of dispersion in TE across trains could help characterise overall temporal complexity of the motif. Similarly, DE takes the absolute differences between consecutive peak amplitudes within trains and adds them together:

$$\mathrm{DE} = \sum_{i=2}^{n} |A_i - A_{i-1}| \tag{4}$$

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where A_i is the amplitude of the i-th peak relative to the maximum amplitude in the Wave, and n is the total number of peaks in the train. In Figure 4, the first train is the faintest in the motif, with the lower DE value and the highest TE value, conveying potentially relevant information about the identity of the signaller.

Both song statistics applications return a set of tables, which can be downloaded as a single Excel Workbook, including summary, motif, train, peak (only in LQ) and the parameters used. TE and DE are reported in the 'Train_Data' table from the song_stats_lq app. All these acoustic metrics could be used in machine learning algorithms to develop detectors and to characterise variability from within individuals to populations. The summary table has 40 columns, including Number of motifs, Pattern Complexity Index (PCI), Duty Cycle, Motif Duration (s), Trains per Motif, Train rate (Hz), Gap duration (s), Entropy, Peak Frequency (kHz), Low Frequency (kHz), High Frequency (kHz), Bandwidth (kHz), Spectral Excursion, Spectral Energy, Spectral Entropy, and Spectral Flatness.

5 | VISUALISATION

Rthoptera offers several apps and functions designed for standardised plotting with presets that produce consistent and reproducible graphs:

- *Oscillogram*: This essential plot in insect bioacoustics can be generated in different ways in the package:
  - oscillogram_zoom: This app allows the user to stack up to two zoomed waveforms below the main oscillogram.
  - oscillogram_ggplot() and oscillogram_plotly(): These functions create a classic oscillogram, static and interactive (HTML), respectively.
  - multioscillo: This app allows stacking oscillograms of different Wave objects, facilitating visual comparison between species or different calls of a single species.
  - colorcode_oscillogram: This app allows colour-coding selections of a spectrogram to, for example, highlight different individuals (e.g., male and female duet).
- *Spectrogram*: Researchers in insect bioacoustics often struggle to find the optimal parameters for a clear spectrogram with an acceptable

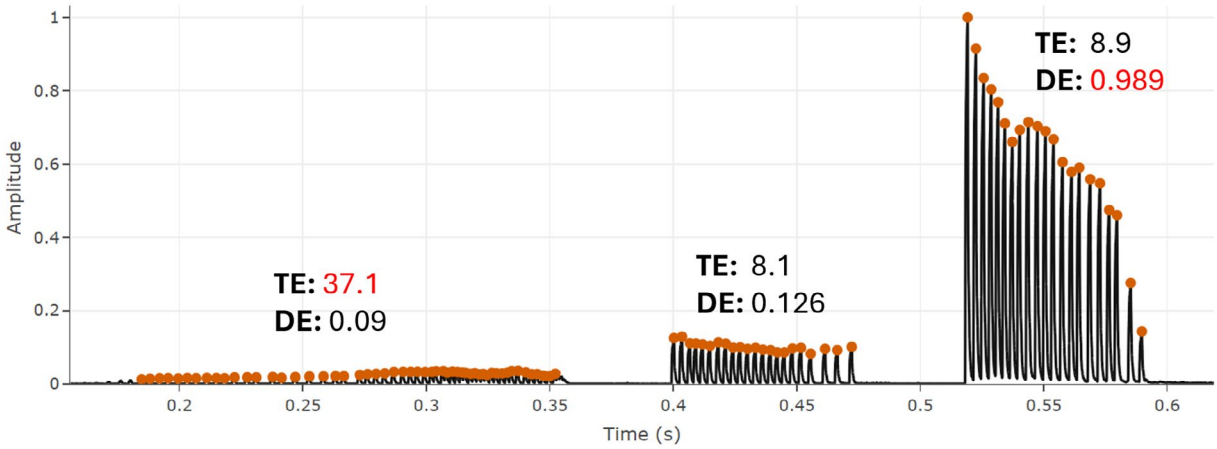

FIGURE 4 Temporal and Dynamic Excursion indices values of a *Coryphoda albidicollis* obtained with the song_stats_lq app. Extreme values for each index in red.

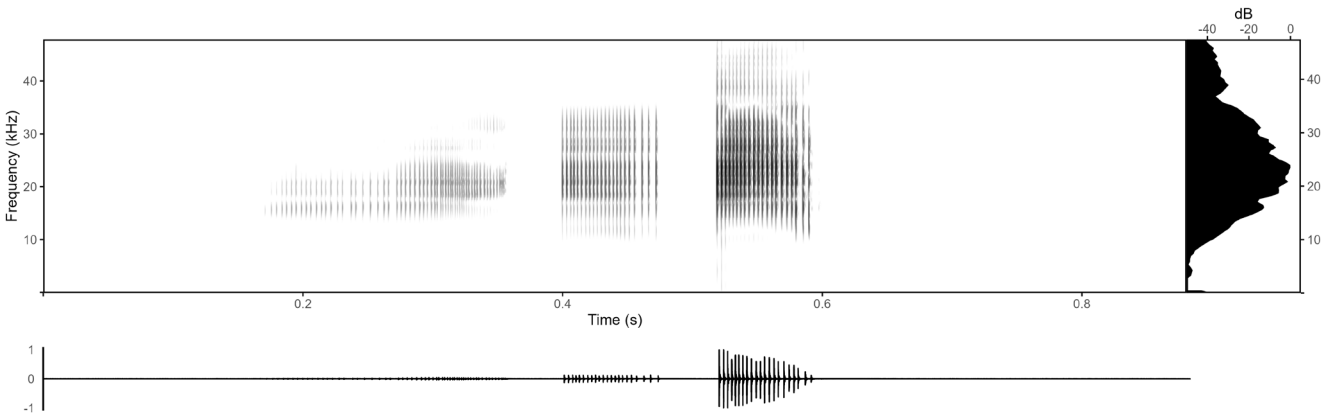

FIGURE 5 Standard multiplot of a calling song of *Coryphoda albidicollis* obtained with one click in the multi_plot app.

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trade-off between spectral and temporal resolutions. The spectrogram app solves this by automatically selecting a window length (WL), accounting for the sampling rate and duration of the Wave:

$$WL = SR \times \sqrt{D} \times k \quad (5)$$

where SR is the sampling rate, D is the duration, and k is a constant (20×10^{-4}). Additionally, the app includes a 'noise floor' parameter, clipping the cells below this threshold (in dB Full Scale) to remove background noise. Optionally, a Mean Power Spectrum (dB or linear scale) can be added on the right side, sharing the *Y*-axis.

- *Multi-plot*. The multi_plot app (Figure 5) allows users to visualise multiple dimensions of a sound simultaneously by grouping a standard spectrogram sharing the *Y*-axis with a mean power spectrum and the *X*-axis with an oscillogram, all in one click when accepting the defaults.
- *Power Spectrum*. Three apps and two regular functions work with this plot type:
  - spectrum. Generates a customisable Power Spectrum (Figure 6).
  - multi_meanspectra_plotly. Allows users to select different sections from an oscillogram and add them to an interactive Mean Power Spectrum Plot, where each selection is automatically

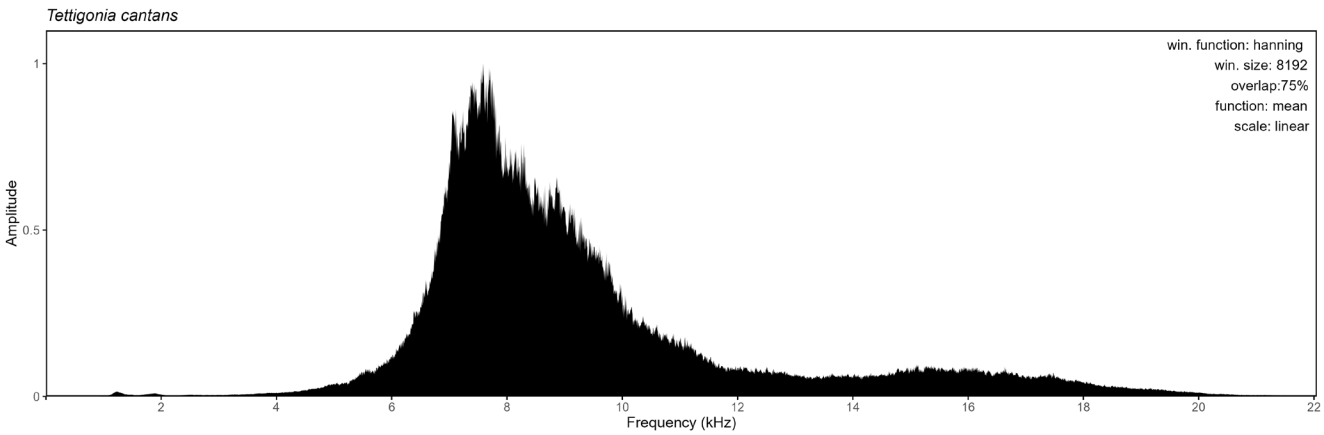

FIGURE 6 Mean Power Spectrum of a calling song of *Tettigonia cantans* from the spectrum app. The parameters in the plot can be toggled off.

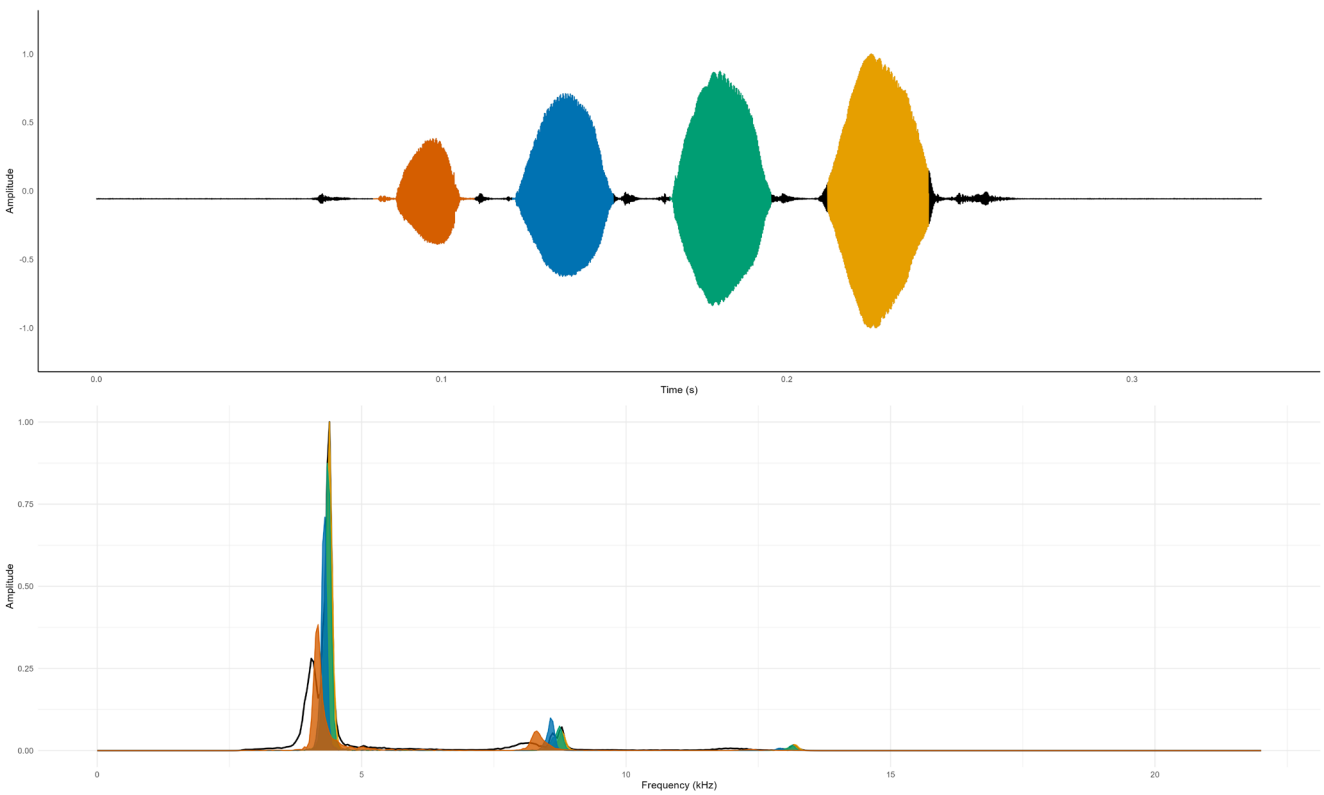

FIGURE 7 Output of the multi_meanspectra_static app showing pulse selections from *Gryllus campestris*.

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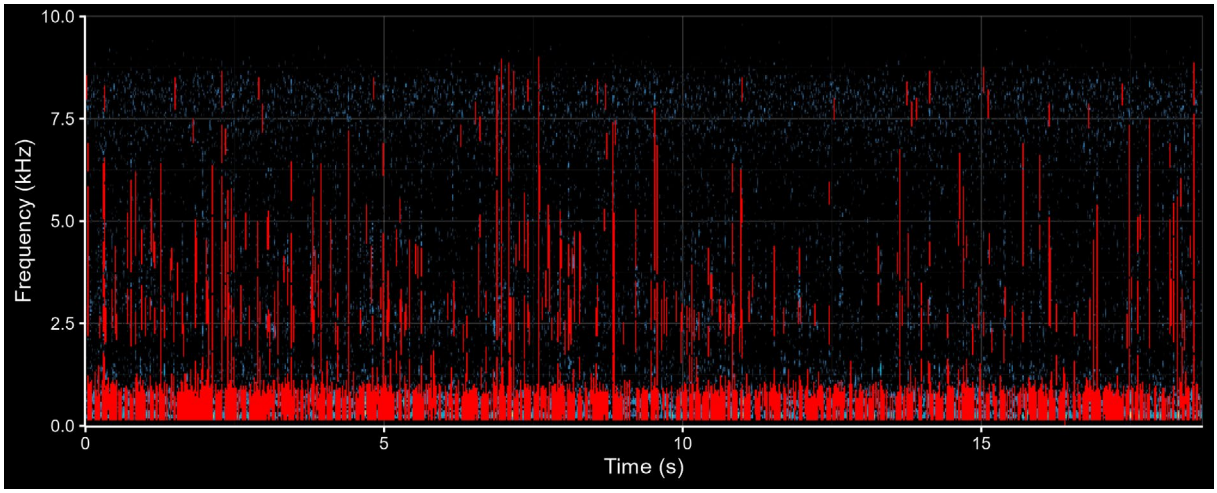

FIGURE 8 Output of the broadband_activity() function using a recording of *Sitophilus zeamais* feeding on maize. The 'clicks' produced by the feeding activity are highlighted in red, while background noise is in light blue.

colour-coded and added to the previous one. The multi_mean-spectra_ggplot app produces a static version (Figure 7), better suited for printed publications.

- spectrum_ggplot() and spectrum_ggplotly() both offer a plot with overlayed linear and dB mean power spectra for static and interactive versions, respectively.

6 | POTENTIAL FOR PEST MONITORING

The broadband_activity() function (Figure 8) calculates the Broadband Activity Index (BBAI) which detects "clicks" (i.e broadband, transient sounds) in a (row-wise) noise-reduced spectrogram, extracting summary statistics including number of clicks, click height and click spectral centroid (kHz). The BBAI corresponds to the proportion of elements in the spectrogram matrix (or cells in the spectrogram) which are classified as part of a click. Originally developed for analysing soundscape recordings to detect noise sources, it shows potential for monitoring non-stridulating insect pests in stored food or wood. Unlike the other measurement tools in *Rthoptera*, this function can discriminate the incidental sound patterns even in challenging recording conditions. Note that the 'spectrogram' argument is set to FALSE by default, so users should set it to TRUE to visualise the plot.

7 | CONCLUSIONS

Insects are important indicators of ecosystem health and agricultural safety, with many species producing sound in terrestrial and aquatic habitats. *Rthoptera* offers an expedited and standardised workflow for insect bioacoustic analysis, facilitating the characterisation of 'acoustic types' for conservation, taxonomic, behavioural, and evolutionary studies. Our approach for signal delimitation in broadband stridulations and the new metrics for temporal and dynamic variability can enrich the bioacoustics toolbox of entomologists by providing detailed automated analyses of small structures. We hope that our package encourages researchers with any degree of expertise to conduct bioacoustics research on insects. Our team welcomes contributions through the Issues section on GitHub: https://github.com/naturewaves/Rthoptera/issues.

AUTHOR CONTRIBUTIONS

Francisco Rivas conceived the package, developed the code and wrote the manuscript. Cesare Brizio co-designed the functionalities of the Shiny applications, tested the functions and wrote the manuscript. Filippo M. Buzzetti provided expert feedback at each stage of development. Bryan Pijanowski provided consistent guidance and funding throughout the process. All co-authors contributed to the manuscript and approved its final version.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

Data available via https://doi.org/10.5281/zenodo.15176395 (Rivas, 2025).

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